08-21 第一篇

Ocean Energy

The ocean covers more than 70% of our planet and holds enormous potential for clean energy. Waves, tides, and even the difference in temperature between warm surface water and cold deep water can be used to generate electricity. [1] As the world looks for ways to reduce carbon emissions, ocean energy is drawing more attention as a renewable and largely untapped resource. Understanding it helps us imagine a greener future.

Wave energy is one of the most promising options. Waves are caused by wind blowing across the sea, and they carry a tremendous amount of power. [2] Special devices can capture this motion and turn it into electricity. Unlike solar panels, wave generators can work day and night. Some coastal countries are already testing wave farms, and the technology is improving every year. What seems like a simple up-and-down movement hides great energy.

Tidal energy uses the rise and fall of the sea level. Twice each day, the tide comes in and goes out, driven by the gravity of the moon and the sun. [3] Engineers build turbines in places where the water flows strongly, and the moving water spins the turbines to produce power. Tidal power is predictable because we know exactly when high and low tide will occur. This reliability makes it attractive for planning energy supply.

Ocean thermal energy conversion, or OTEC, uses the temperature difference between warm surface water and cold deep water. [4] This technology works best in tropical regions where the difference is large. Although OTEC plants are expensive to build, they can provide steady power with little pollution. Research continues to find cheaper and more efficient designs. The ocean's natural temperature gradient may become an important energy source in the decades to come.

Ocean energy is still in early development. [5] More investment, better technology, and stricter environmental rules will help it grow in a sustainable way. Students who study science and engineering today may one day help design the ocean energy systems of the future. The sea that has long inspired explorers may soon power our homes and cities.

选项

A. That is why many nations are now investing in research and pilot projects. B. The constant movement of waves provides a steady source of power. C. In such locations, the daily change in water level can be several meters. D. Warm water is used to evaporate a special fluid, which then drives a turbine to generate electricity. E. With the

right support, it could play a major role in our clean energy mix. F. For example, some floating devices rise and fall with each wave to capture energy.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前介绍海洋的多种能源形式及清洁潜力，空后提到"世界寻找减排途径、海洋能源作为可再生且未充分利用的资源受到关注"。选项 A"这就是为什么许多国家正在投资研究和试点项目"承接海洋能源的潜力，自然引出后文对其作为替代能源的重视。
- **[2] B**：
  - **线索**：空前说明波浪由风引起、蕴含巨大能量，空后描述"特殊装置可捕捉运动并转化为电力"。选项 B"波浪的持续运动提供稳定的电力来源"与前文" tremendous amount of power"呼应，与后文"全天候工作、不同于太阳能"形成逻辑支撑，解释波浪发电的优势。
- **[3] C**：
  - **线索**：空前介绍潮汐的日两次涨落及日月引力驱动，空后说"工程师在潮流强处建涡轮、水流驱动发电"。选项 C"在这些位置，每日水位变化可达数米"与前文"潮汐涨落"呼应，量化潮差，解释何以在特定地点建涡轮，与后文"潮汐可预测、便于能源规划"形成递进。
- **[4] D**：
  - **线索**：空前介绍 OTEC 利用表层温水与深层冷水的温差，空后提及"该技术在温差大的热带最有效"。选项 D"温水用于蒸发特殊流体，流体再驱动涡轮发电"具体说明 OTEC 的运作原理，与前文"温差"形成技术链条，与后文"建造成本高、研究改进"自然衔接。
- **[5] E**：
  - **线索**：空前指出海洋能源仍处于早期发展，空后列举"更多投资、更好技术、更严环保规则"及学生未来角色。选项 E"在适当支持下，它可在清洁能源组合中扮演重要角色"与前文"早期发展"形成转折，与后文"投资、技术、规则"及"未来为家庭城市供电"形成呼应。
- **干扰项分析**：

- **F. For example, some floating devices rise and fall with each wave to capture energy**：以 For example 开头的典型干扰项。与第二段波浪发电高度相关，易被误选为 [2]。但 F 是单一装置类型的举例，而 [2] 处需要的是对波浪发电原理或优势的概括；且 F 的"floating devices"层次过细，与"持续运动、稳定供电"的主旨有偏差，属举例型干扰。